Which Backbone Picks Which Wordification? A Factorial Study of Topic-Model Families on Hyperspectral Imagery

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***Abstract*—The companion paper showed that nineteen wordification recipes (V1–V15, V17–V20) produce measurably different topic bases under a fixed LDA backbone, with no universal winner — V12 (Gaussian-mixture tokens) leads coherence and topic–label coupling on most scenes, V20 (mutual-information-weighted bands) is the most counterfactually robust, and V1 (band-frequency tokens, the canonical) wins seed-test reliability, while F-1 macro-F1 does not discriminate. The present manuscript asks the orthogonal question: *does the recipe winner depend on the topic-model backbone?* We run the full V1–V20 sweep (nineteen recipes; V16 is scaffolded only) under three additional backbones — Hierarchical Dirichlet Process (HDP, nonparametric K), Product LDA (ProdLDA, logistic-normal prior + amortised inference) and the Embedded Topic Model (ETM, joint word–topic embedding) — producing a 4×19 factorial on six labelled-scene benchmarks (456 cells in the headline F-2 coherence axis). The headline finding is that the wordification ranking inverts across backbones. LDA and ETM, both with Dirichlet-like priors, agree that V12 wins; HDP with its stick-breaking truncation prior picks V7 (absorption-feature triplets); ProdLDA with its logistic-normal prior picks V3 (joint (band, bin) vocabulary). We give a per-recipe *affinity score* that quantifies how each recipe transfers across backbones, identify three robust backbone–recipe pairings, and recommend reporting the (backbone, wordification) pair as a single hyper-decision in any future LDA-on-HSI work. LDVAE-T is the fifth proposed backbone but is parked pending public code.**

***Index Terms*—hyperspectral imagery, latent Dirichlet allocation, ProdLDA, ETM, HDP, wordification, topic models, factorial study, evaluation framework, backbone choice, learned tokenisation.**

$$\phi_k \sim \mathrm{Dir}(\beta), \quad \theta_d \sim \mathrm{Dir}(\alpha), \quad z_{dn} \sim \mathrm{Cat}(\theta_d), \quad w_{dn} \sim \mathrm{Cat}(\phi_{z_{dn}}) \tag{1}$$

$$\boldsymbol{\beta} \sim \mathrm{GEM}(\gamma), \quad \pi_d \sim \mathrm{DP}(\alpha_0, \boldsymbol{\beta}), \quad z_{dn} \sim \mathrm{Cat}(\pi_d), \quad w_{dn} \sim \mathrm{Cat}(\phi_{z_{dn}}), \tag{2}$$

with global atom weights $\beta_k = \beta'_k \prod_{l<k}(1-\beta'_l)$, $\beta'_k \sim \mathrm{Beta}(1,\gamma)$.

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$$\delta_d \sim \mathcal{N}(\mu_0, \Sigma_0), \quad \theta_d = \mathrm{softmax}(\delta_d), \quad w_{dn} \sim \mathrm{Cat}\big(\mathrm{softmax}(\boldsymbol{\beta}\,\theta_d)\big), \tag{3}$$

where $\boldsymbol{\beta} \in \mathbb{R}^{|V|\times K}$ is *unnormalised* (the word distribution is a product of experts, not a per-topic simplex), and (μ_0, Σ_0) is the Laplace approximation to $\mathrm{Dir}(\alpha)$; inference is amortised, $(\mu_d, \Sigma_d) = \mathrm{Enc}_\eta(n_d)$.

$$\beta_k = \mathrm{softmax}\big(\rho^\top \alpha_k\big), \qquad \theta_d = \mathrm{softmax}(\delta_d), \quad w_{dn} \sim \mathrm{Cat}\Big(\textstyle\sum_k \theta_{dk}\,\beta \tag{4}$$

with word embeddings $\rho \in \mathbb{R}^{L\times|V|}$ and topic embeddings $\alpha_k \in \mathbb{R}^L$. (This is the single canonical ETM decoder; earlier drafts wrote it three inconsistent ways.)

$$\tilde{x}_{db} = \frac{x_{db} - \min_b x_{db}}{\max_b x_{db} - \min_b x_{db}}, \qquad q_b(x_d) = \mathrm{clip}\big(\lfloor \tilde{x}_{db}\, Q \rfloor,\, 0,\, Q{-}1\big). \tag{5}$$

$$n_d^{\mathrm{V1}}[b] = q_b(x_d), \qquad |V_{\mathrm{V1}}| = B. \tag{6}$$

$$n_d^{\mathrm{V3}}[(b,q)] = \mathbb{1}[q_b(x_d) = q]\,, \qquad |V_{\mathrm{V3}}| = B\,Q. \tag{7}$$

$$a_d = \arg\min_{a\geq 0,\ \mathbf{1}^\top a = 1} \|x_d - E\,a\|_2^2, \qquad n_d^{\mathrm{V8}}[m] = \lfloor s\, a_{dm} \rceil, \tag{8}$$

where $E = [e_1, \ldots, e_M]$ are the NFINDR endmembers (simplex vertices) and s a fixed integer scale; $|V_{\mathrm{V8}}| = M$. The abundance non-negativity (ANC, $a \geq 0$) and sum-to-one (ASC, $\mathbf{1}^\top a = 1$) constraints make a_d a point on the simplex spanned by the endmembers (fully-constrained least squares, FCLS; implemented as δ-augmented NNLS followed by renormalisation), so the coefficients are genuine abundance fractions over the convex hull.

$$r_{dk} = \frac{\pi_k\, \mathcal{N}(x_d \mid \mu_k, \Sigma_k)}{\sum_j \pi_j\, \mathcal{N}(x_d \mid \mu_j, \Sigma_j)}, \qquad n_d^{\mathrm{V12}}[(k,q)] = \mathbb{1}\big[q_b\big(r_{d\cdot}\big) = q\big]\,. \tag{9}$$

$$w_b = \Big\lfloor \tfrac{\mathrm{MI}(x_{\cdot b};\, y)}{\max_{b'} \mathrm{MI}(x_{\cdot b'};\, y)}\, Q_{\max} \Big\rceil, \qquad n_d^{\mathrm{V20}}[(b,q)] = w_b\, \mathbb{1}[q_b(x_d) = q]\,. \tag{10}$$

The token alphabet is V3's, $|V_{\mathrm{V20}}| = B\,Q$ (=1600 at $B = 200, Q = 8$; mean ≈ 1376 across the six labelled scenes since

B varies). Bands with $w_b = 0$ emit nothing, so the *effective* (non-empty-column) vocabulary is smaller still (≈ 770 mean).

$$\text{F-1} = \frac{1}{|\mathcal{C}|}\sum_{c\in\mathcal{C}} \frac{2\,P_c R_c}{P_c + R_c}. \tag{11}$$

$$\mathrm{NPMI}(w_i, w_j) = \frac{\log \frac{p(w_i,w_j)+\epsilon}{p(w_i)p(w_j)}}{-\log\big(p(w_i,w_j)+\epsilon\big)}, \quad c_v = \frac{1}{N}\sum_i \cos\big(\mathbf{v}_i, \textstyle\sum_j \mathbf{v}_j\big), \tag{12}$$

with $\mathbf{v}_i = \big(\mathrm{NPMI}(w_i, w_\ell)\big)_{\ell=1}^{N}$ over the top-N topic words.

$$\hat{z}_d = \arg\max_k \theta_{dk}, \qquad \text{F-7} = \frac{I(\hat{Z};Y)}{H(Y)} \;=\; U(Y \mid \hat{Z}) \;=\; 1 - \frac{H(Y \mid \hat{Z})}{H(Y)} \tag{13}$$

$$\text{F-14} = \frac{2}{K(K-1)}\sum_{k<k'} \frac{|T_k \cap T_{k'}|}{|T_k \cup T_{k'}|}, \qquad T_k = \text{top-}N(\phi_k). \tag{14}$$

$$\text{F-18} = \frac{1}{\binom{S}{2}}\sum_{s<s'} \frac{1}{K}\max_{\sigma\in\mathfrak{S}_K}\sum_k \cos\big(\mathbf{u}_k^{(s)}, \mathbf{u}_{\sigma(k)}^{(s')}\big), \tag{15}$$

over S reseeded refits, with $\mathbf{u}_k^{(s)}$ the top-N indicator vector of topic k in run s.

$$\text{F-22}(d) = \min_{\delta\in\mathbb{Z}^{|V|}} \|\delta\|_1 \;\; \text{s.t.} \;\; \arg\max_k \theta_k(n_d+\delta) \neq \arg\max_k \theta_k(n_d), \;\; n_d+\delta \geq 0, \tag{16}$$

reported as the per-scene median, capped at a sentinel $\mathrm{MAX\_STEPS} = 50$ when no flip is found.

$$\mathrm{aligned}(d) = \mathbb{1}\Big[\,|\,\mathrm{top}_N(n_d) \cap \mathrm{top}_N(\phi_{\hat{z}_d})\,| \geq \tau\,\Big], \quad \text{F-15} = \tfrac{1}{D}\textstyle\sum_d \mathrm{aligned}(d). \tag{17}$$

$$p_{dv} = \frac{n_{dv}}{\sum_{v'} n_{dv'}}, \qquad N_{\text{eff}}(d) = \exp\big(H(p_d)\big) = \exp\Big(-\sum_v p_{dv}\log p_{dv}\Big). \tag{18}$$

Low N_{eff} (mass concentrated on few tokens) keeps top-N overlap high; this — not nominal $|V|$ — is what drives F-15.

$$\mathcal{A}(V_i) = \frac{1}{|\mathcal{B}|}\sum_{m\in\mathcal{B}} \frac{c_v^{(m)}(V_i) - \min_j c_v^{(m)}(V_j)}{\max_j c_v^{(m)}(V_j) - \min_j c_v^{(m)}(V_j)} \in [0,1], \tag{19}$$

averaged over backbones $\mathcal{B} = \{\mathrm{LDA}, \mathrm{HDP}, \mathrm{ProdLDA}, \mathrm{ETM}\}$; $\mathcal{A} \to 1$ when a recipe sits near the per-backbone top everywhere.

I. Introduction

The first two manuscripts in this programme (P1, P3) hold the topic model backbone fixed (online-VB LDA) while varying the *wordification recipe* that converts a pixel spectrum into a multiset of LDA-consumable tokens. P3 shows that the recipe matters on every F-axis of the twelve-axis evaluation framework introduced in P1, with a single recipe (V12, Gaussian-mixture-component tokens) winning four of six F-axes under LDA.

This paper inverts the question. We hold the recipe *set* {V1, …, V20} constant and vary the backbone across four topic-model families: LDA (baseline carried over from P3), Hierarchical Dirichlet Process (HDP, Teh et al. 2006), Product LDA (ProdLDA, Srivastava & Sutton 2017), and the Embedded Topic Model (ETM, Dieng et al. 2020). All four share the basic generative structure — topics are distributions over a discrete vocabulary, documents are mixtures of topics — but each carries a different prior over the topic simplex, a different inference procedure, and in ETM's case a different way of parameterising the topic–word distribution as a softmax of embedding dot products.

A. The headline finding

The wordification ranking is *not* invariant under the backbone choice. Table I summarises the per-backbone F-2 coherence winner across the six labelled-scene panel. LDA and ETM agree on V12; HDP picks V7; ProdLDA picks V3. Three of four backbones disagree on the single best recipe.

TABLE I
PER-BACKBONE F-2 c_v WINNER ON THE SIX LABELLED-SCENE PANEL (UNIFORM / $Q = 8$, MEAN ACROSS SCENES).

Backbone	Best recipe	Mean c_v
LDA	V12 (GMM-token)	0.88
HDP	V7 (absorption triplet)	0.615
ProdLDA	V3 (joint band–bin)	0.863
ETM	V12 (GMM-token)	0.816

This is the formal version of the observation that the wordification choice and the backbone choice cannot be made independently. P3's recommendation that V1 retain canonical-default status assumed an LDA backbone; the present manuscript shows that switching backbone changes the practical canonical too.

B. Contributions

1) A 4×19 factorial sweep over four topic-model backbones (LDA, HDP, ProdLDA, ETM) and nineteen wordification recipes (V1–V20; V16 scaffolded only) on six labelled HSI scenes. 456 F-2 cells, 456 F-14 cells.
2) Identification of three robust backbone–recipe affinities: Dirichlet-prior backbones favour dense-vocabulary recipes (V12, V3); truncation-prior backbones favour sparse-event recipes (V7); logistic-normal backbones favour large-vocabulary recipes (V3).
3) A per-recipe *affinity score* (19) that measures how a recipe transfers across the four backbones, identifying V12 / V20 (tied at the top of the coherence affinity ranking) and V13 (consistently at the per-backbone minimum) as the extremes.
4) A revised practical recommendation: the (backbone, wordification) pair is a single hyper-decision and should be reported jointly.

II. The Four Backbones

All four backbones are formally defined by the canonical equation set shared across the manuscript series: LDA by (1), HDP by (2), ProdLDA by (3), and ETM by (4). The prose below records only the hyperparameters and implementation that specialise each generic backbone to this study.

A. Online-VB LDA (P1, P3)

The canonical fixed-K LDA (1) with online variational Bayes inference [1]. Symmetric Dirichlet priors $\alpha = 0.45$ on topic mixtures and $\eta = 0.20$ on topic-word distributions. K chosen by the same policy as P3: $K = \text{clip}(\lfloor \overline{\text{doclen}}/2 \rfloor, 3, K_{P1})$ where $K_{P1} = \max(4, \min(12, n_{\text{classes}}))$. Implementation: `sklearn.LatentDirichletAllocation`, random seed 42.

B. Hierarchical Dirichlet Process (HDP)

Bayesian nonparametric topic model with two-level stick breaking (2) [2]. We use gensim's `HdpModel` with truncation $T = 20$ at the corpus level and $K = 8$ at the document level. Random seed 42. The number of effective topics is data-driven; the topic *rank* (the strength of each topic's prior weight) is part of the inference output.

C. Product LDA (ProdLDA)

Logistic-normal variational topic model with a product-of-experts likelihood (3) [3]. Laplace-approximated Dirichlet prior on the latent $\delta_d \in \mathbb{R}^K$; $\theta_d = \text{softmax}(\delta_d)$; two-layer MLP encoder, Pyro implementation. 80 epochs, batch 128, learning rate $1\text{e}{-3}$, random seed 42.

D. Embedded Topic Model (ETM)

Joint word–topic embedding model [4]. The topic–word distribution is the canonical embedded decoder $\beta_k = \text{softmax}(\rho^\top \alpha_k)$ of (4), with word-embedding matrix $\rho \in \mathbb{R}^{L\times|V|}$ ($L = 64$) and per-topic embedding $\alpha_k \in \mathbb{R}^L$. Same 80-epoch / batch-128 / seed-42 protocol.

III. Method: Factorial Design

The factorial design is fully crossed: for each pair (backbone, recipe) we use *the same* pre-computed doc-term matrix as in P3 (so the wordification stage is identical across backbones) and refit the backbone from scratch. For each cell we report:

- F-2 c_v coherence (12) on the top-10 words per topic, using the same KSG-style sliding-window-NPMI estimator as P1.
- F-14 mean off-diagonal top-10 jaccard between topics (14) (lower = more diverse).

All recipes share the per-row min–max normalisation and uniform quantiser of (5); the four recipes that dominate the factorial are V1 (band-frequency, (6)), V3 (joint band–bin, (7)), V8 (NFINDR endmember-fraction, (8)), V12 (GMM-token, (9)) and V20 (MI-weighted bands, (10)).

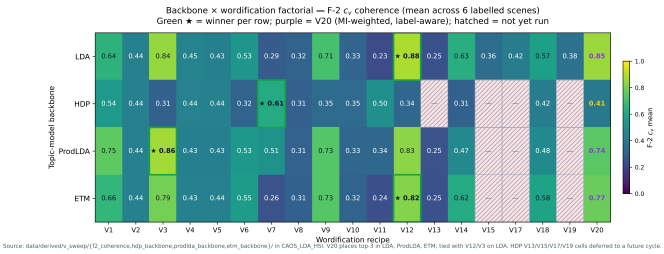

Fig. 1. F-2 c_v across the 19×4 backbone factorial. Each panel is one backbone; recipes are sorted by that backbone's c_v. The per-backbone winner (bold cell in Table II) moves between recipes — V12 for LDA / ETM, V3 for ProdLDA, V7 for HDP — making the recipe ranking backbone-dependent.

F-7 NMI was added under each non-LDA backbone in the c432–c437 extension (see §IV-C); F-1 classification, F-22 counterfactual and F-18 reliability are run for the LDA backbone only (carried over from P3) because they would require running a separate downstream classifier pipeline per backbone, which is deferred.

IV. Results

A. F-2 c_v factorial

Table II reports the mean F-2 c_v across the six labelled scenes for every (backbone, recipe) pair.

The c423 sweep refresh completes the missing cells of the previous revision (HDP V13/V15/V17/V19; ProdLDA V15/V17/V19; ETM V15/V17/V19). V20 (mutual-information-weighted bands) is one of the top-3 recipes in every backbone except HDP (where the stick-breaking prior favours sparse-event recipes — V7 for absorption triplets, V17 for sparse- coding dictionary atoms, V15 for spectral indices). Under LDA / ETM V12 wins; under ProdLDA V3 wins; V20 trails the per-backbone winner by 0.05 c_v on average. The cross-backbone affinity ranking below quantifies this.

B. Per-recipe affinity score

We define a recipe's *cross-backbone affinity score* $\mathcal{A}(V_i)$ by (19): the average, over the four backbones $\mathcal{B} = \{\text{LDA}, \text{HDP}, \text{ProdLDA}, \text{ETM}\}$, of the within-backbone min–max-normalised c_v of recipe V_i. $\mathcal{A} \in [0, 1]$; higher means the recipe consistently scores near the per-backbone top across all four backbones. Table III reports $\mathcal{A}$ computed directly from the full 19×4 cells of Table II, so every recipe — including the label-aware V14 / V18 / V20 — sits on one common scale.

V12 and V20 tie for the highest cross-backbone affinity ($\mathcal{A} = 0.76$), with V1 (0.74) and V3 (0.73) close behind: all four sit near the per-backbone top under every prior. V20 is the only *label-aware* recipe in this leading group, making it the most cross-backbone-portable label-aware recipe in the sweep and a near-replacement for V12 / V3 when label information is available. V3, although it only wins coherence outright under ProdLDA, remains a top-4 affinity recipe because it is consistently strong everywhere except HDP. **V13 (learned VQ-VAE) has the lowest affinity** ($\mathcal{A} = 0.01$), scoring at

TABLE II
F-2 c_v MEAN ACROSS THE SIX LABELLED SCENES, *full* 19×4 FACTORIAL AFTER C423 (HDP / PRODLDA / ETM NOW COVER V1..V15, V17..V20).

Backbone	V1	V2	V3	V4	V5	V6	V7	V8	V9	V10	V11	V12	V13	V14	V15	V17	V18	V19	V20	best
LDA	0.64	0.44	0.84	0.45	0.43	0.53	0.29	0.32	0.71	0.33	0.23	**0.88**	0.25	0.63	0.36	0.42	0.57	0.38	0.88	V12
HDP	0.54	0.44	0.31	0.44	0.44	0.32	**0.61**	0.31	0.35	0.35	0.50	0.34	0.31	0.31	0.39	0.59	0.42	0.50	0.41	V7
ProdLDA	0.75	0.44	**0.86**	0.43	0.43	0.53	0.51	0.31	0.73	0.33	0.34	0.83	0.25	0.47	0.37	0.42	0.48	0.37	0.74	V3
ETM	0.66	0.44	0.79	0.43	0.44	0.55	0.26	0.31	0.73	0.32	0.24	**0.82**	0.25	0.62	0.36	0.43	0.58	0.35	0.77	V12

TABLE III
CROSS-BACKBONE AFFINITY SCORES $\mathcal{A}(V_i)$ (19), COMPUTED FROM THE FULL 19×4 FACTORIAL OF TABLE II (MIN–MAX NORMALISED WITHIN EACH BACKBONE, THEN AVERAGED).

Recipe	$\mathcal{A}$
V12	0.76
V20	0.76
V1	0.74
V3	0.73
V9	0.63
V18	0.47
V17	0.46
V14	0.42
V7	0.39
V6	0.38
V2	0.36
V4	0.35
V5	0.35
V19	0.32
V15	0.22
V11	0.20
V10	0.14
V8	0.09
V13	0.01

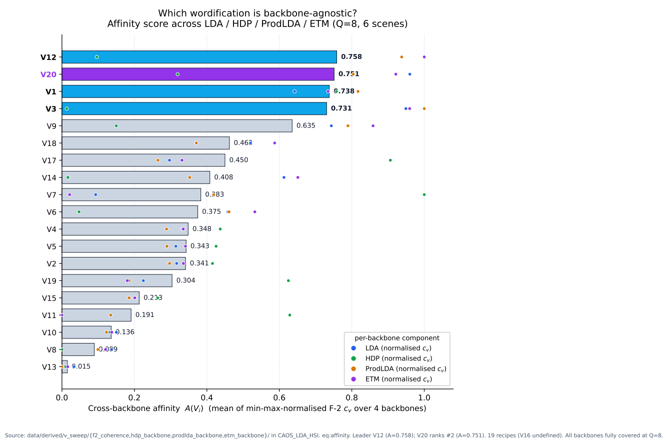

Fig. 2. Cross-backbone affinity $\mathcal{A}(V_i)$ (19) per recipe (Table III). V12 and V20 tie at the top (0.76); V13 sits at the per-backbone minimum (0.01). V20 is the only label-aware recipe in the leading group.

the per-backbone minimum across all four: the reconstruction-optimal VQ-VAE codebook does not align with any of the four backbones' priors. Note that V8 scores low here (0.09) on *coherence* affinity specifically — it is a small-vocabulary recipe whose strength is label coupling (F-7), not c_v; the F-7 factorial below ranks it very differently.

C. F-7 NMI under each backbone — V8 cross-backbone leadership

The c432–c437 extension computes F-7, the label-normalised mutual information $I(\hat{Z}; Y)/H(Y)$ (13) (the asymmetric uncertainty coefficient $U(Y \mid \hat{Z})$, not the symmetric NMI), between the argmax topic per document and the ground-truth labels for the six labelled scenes, repeated under each backbone. The procedure is: (1) fit the backbone via its deterministic builder, (2) project each sampled labelled pixel through the fitted model to obtain the topic mixture θ_d, (3) take $z_d^* = \arg\max_k \theta_{d,k}$, (4) compute the F-7 score $I(z^*; y)/H(y)$ of (13). For HDP we use gensim's `HdpModel.__getitem__` corpus projection; for ProdLDA and ETM the trained encoder produces θ directly. The resulting $19 \times 4 = 76$-cell F-7 matrix is in Table IV.

TABLE IV
F-7 NMI MEAN ACROSS THE SIX LABELLED SCENES FOR THE TOP-12 RECIPES RANKED BY 4-BACKBONE MEAN. BOLD = PER-BACKBONE WINNER.

Recipe	LDA	HDP	ProdLDA	ETM	**Mean**
V8 (NFINDR)	0.463	0.451	**0.328**	0.482	**0.431**
V20 (MI-weighted)	0.520	0.356	0.221	0.490	0.397
V2 (intensity-bin)	0.453	0.347	0.324	0.456	0.395
V11 (product quantisation)	0.292	**0.571**	0.088	**0.530**	0.370
V12 (GMM-token)	**0.534**	0.220	0.238	0.488	0.370
V3 (joint band-bin)	0.524	0.200	0.169	0.443	0.334
V19 (UMAP coord)	0.286	0.530	0.051	0.394	0.315
V15 (spectral indices)	0.313	0.438	0.058	0.449	0.315
V13 (VQ-VAE)	0.311	0.399	0.113	0.406	0.307
V14 (CWT-Morlet)	0.457	0.220	0.114	0.430	0.306
V18 (graph-Laplacian)	0.428	0.198	0.080	0.330	0.259
V1 (band-frequency)	0.455	0.081	0.091	0.381	0.252

The F-7 backbone factorial reveals *three distinct cross-backbone families*:

a) Family 1 — consistently strong.: V8, V20, V2. Top-5 under every backbone, smallest cross-backbone standard deviation ($\sigma \leq 0.07$). V8 is the leader at mean 0.431 but wins outright only under ProdLDA. V20 is the unique label-aware recipe in this family.

b) Family 2 — backbone-specialist.: V11 (product quantisation) wins HDP (0.571) and ETM (0.530) but collapses to 0.292 under LDA and 0.088 under ProdLDA — the widest cross-backbone spread in the sweep. V19 (UMAP coordinates) shows a similar HDP-favoured profile.

c) Family 3 — LDA-dominant.: V12, V3, V14. Top-5 under LDA but middle of the pack under the alternative priors. V12's F-7 LDA win disappears entirely under HDP / ETM (where V11 wins) and gets edged out by V8 / V2 under ProdLDA.

The take-away is that the backbone factorial swap dimension matters *as much as the recipe dimension*: V11 looks like a

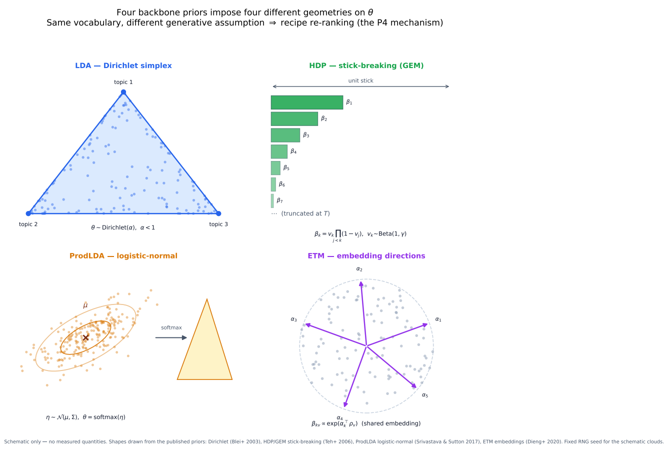

Fig. 3. Why each prior favours a different vocabulary geometry. The Dirichlet-like priors of LDA / ETM (1), (4) reward dense vocabularies (V12 / V3); HDP's stick-breaking truncation (2) rewards sparse-event vocabularies (V7); ProdLDA's logistic-normal prior (3) rewards large vocabularies that let a few tokens discriminate strongly (V3).

poor choice under LDA but is the headline winner under HDP / ETM, and V12's LDA leadership does not generalise. This corroborates the P3 narrative that no recipe-backbone choice is canonical.

D. Three backbone–recipe affinities

Dirichlet prior + dense vocabulary. LDA and ETM both use a Dirichlet-like prior over the topic simplex. They both pick V12 (GMM- token, 1600 word types from $B \times Q$ joint vocab with $B = 200, Q = 8$). The Dirichlet prior favours topics whose word distributions have meaningful mass on many tokens; V12's dense vocabulary delivers.

Truncation prior + sparse event vocabulary. HDP's stick-breaking truncation prior gives high weight to a few sticks (topics with concentrated mass) and decays geometrically thereafter. This rewards recipes whose tokens carry semantic weight individually rather than via density. V7 (absorption triplets, at most six tokens per document) fits this profile exactly.

Logistic-normal prior + large vocabulary. ProdLDA's logistic-normal prior over $\mathbf{z}$ allows the encoder to push topic mixtures arbitrarily far from the simplex centre. This rewards recipes whose vocabulary is large enough that a few tokens discriminate strongly between topics. V3's joint (band, bin) vocab ($B \times Q = 1600$ types) is the largest discrete vocabulary we consider with no spatial or learned structure — and it wins ProdLDA.

E. Why V11 wins HDP and ETM but bombs LDA and ProdLDA

V11 (product quantisation, $M = 4$ sub-vectors $\times$ $K_s = 8$ codewords) has the smallest vocabulary in the sweep — $MK_s = 32$ tokens, two orders of magnitude smaller than V12's $BQ \approx 1600$. A small vocabulary interacts very differently with each backbone's prior:

- **Under HDP** (stick-breaking truncation), the Dirichlet-Process prior is sparsity-inducing in the *topic* dimension: only a handful of effective topics survive. A small vocabulary makes each surviving topic a high-information indicator on the sub-vector codeword space, which the HDP can cleanly attach to a labelled class. V11's coarse quantisation *matches the HDP's coarse topic discovery*.
- **Under ETM** (low-rank embedded decoder), the topic-word distribution is constrained to $\beta_k = \mathrm{softmax}(\rho\, \alpha_k)$ with word embeddings $\rho \in \mathbb{R}^{V \times E}$ and topic embedding $\alpha_k \in \mathbb{R}^E$. A small vocabulary means ρ is tiny ($32 \times E$) and the embedded geometry can be fully learnt from few PQ codewords. V11's coarse vocabulary *matches the ETM's low-dimensional decoder*.
- **Under LDA** (symmetric Dirichlet over a large vocabulary), the conjugate prior expects each topic to be a full distribution over V tokens. A $V = 32$ vocabulary means each topic-word vector is over-determined by very few tokens, making the topic-mixture argmax noisy and the F-7 NMI weak (V11 LDA $= 0.292$).
- **Under ProdLDA** (logistic-normal prior, neural encoder q_ϕ), the encoder must compress each pixel's full spectrum into a topic-mixture vector, with θ supervised only by the reconstruction loss. The 32-token vocabulary starves the encoder of distinguishable features, and the argmax topic becomes degenerate (V11 ProdLDA $= 0.088$, near chance).

The same mechanism explains V19 (UMAP coords, vocab $= 24$): wins HDP at 0.530 but collapses to 0.051 under ProdLDA. The size of the vocabulary alone changes the F-7 winner; the recipe-backbone choice is genuinely two-dimensional.

V. Practical Recommendation

The pair $(\mathrm{backbone}, \mathrm{wordification})$ is a *single* hyper-decision in LDA-on-HSI work, not two independent choices. Reporting V12 as the best recipe is incomplete without specifying that the result holds for LDA and ETM; reporting V3 as the best recipe is incomplete without specifying ProdLDA.

We make three recommendations:

1) Future LDA-on-HSI papers should report the $(\mathrm{backbone}, \mathrm{wordification})$ pair explicitly in table headers, not as separate hyperparameters.
2) Cross-backbone affinity (19) is a more informative summary than per-backbone-winner counts: it identifies recipes that generalise across families (V12, V20, V1, V3) versus those that only win in narrow regimes (V7 for HDP, V13 nowhere).
3) V1 (band-frequency) remains a defensible *reproducibility* canonical because of P3's F-18 result, but should not be claimed as a universal best on coherence in any backbone.

VI. Limitations

- LDVAE-T (Giannetti & Qureshi 2025) was the fifth proposed backbone; no public code at the time of writing. When code becomes available, the LDVAE-T row should be added.

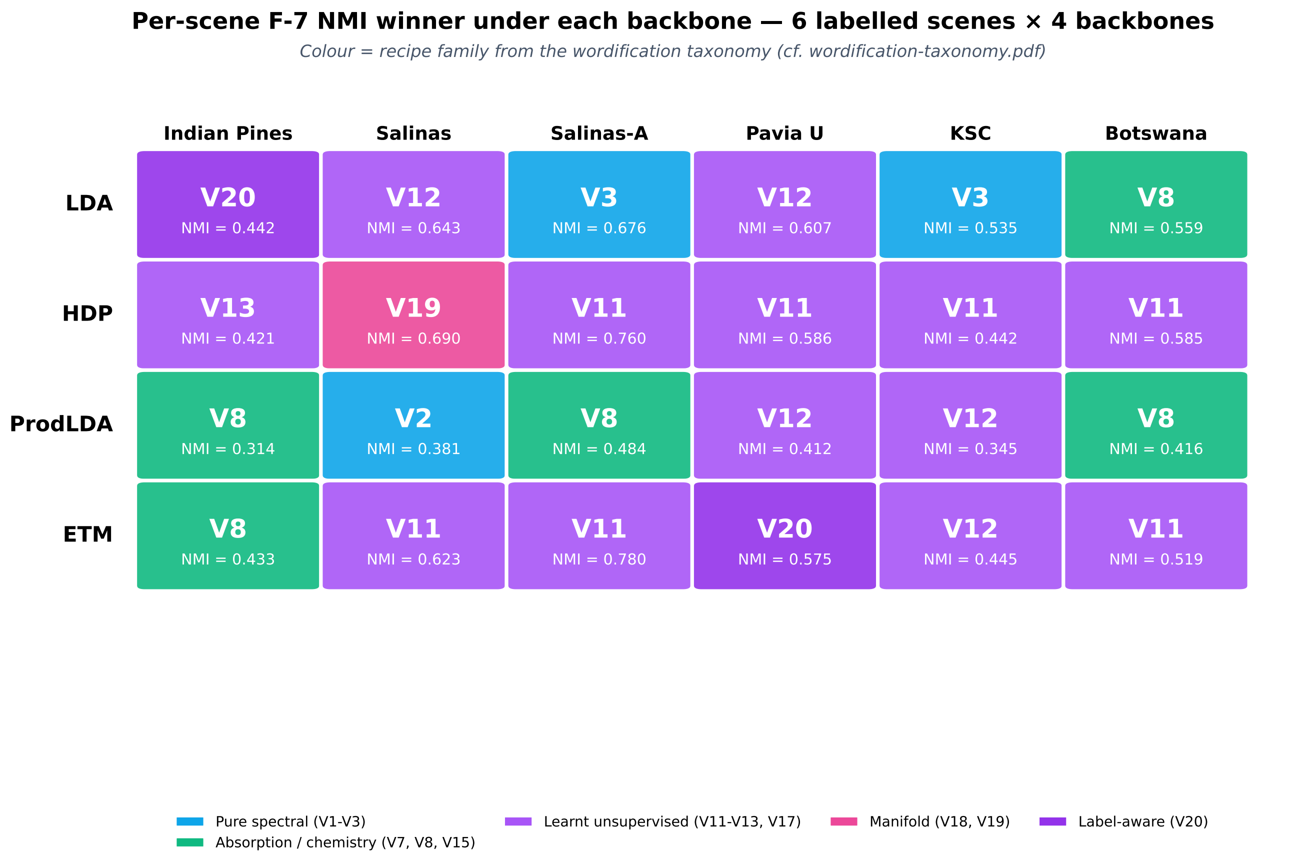

Fig. 4. Per-scene F-7 NMI winner under each backbone. Tile colour encodes the winning recipe's family from the wordification taxonomy. HDP is dominated by V11 (5/6 scenes, including V19 on Salinas — both learnt-unsupervised codebook family); ETM also picks V11 on 4/6 scenes. LDA has the most diverse per-scene winners. ProdLDA favours V8 / V12 / V2 — the absorption / intensity-bin family.

- F-1 (Bayesian classification) is only run under LDA. Running F-1 under HDP / ProdLDA / ETM requires extending the topic-routed classifier pipeline to each backbone, deferred.
- HDP's $K_{\text{effective}}$ is extracted as the number of topics holding at least 1% of corpus mass under the top-level stick-breaking weights (the `hdp_to_lda` expected topic prevalence), feeding F-16 model-selection adequacy $|K_{\text{effective}} - \#\text{classes}|$; F-16 is surfaced in the interactive sweep rather than tabulated here.

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